

The image displays a series of Feynman diagrams representing a scattering process, separated by a vertical dashed red line. The diagrams involve external momenta p_1 and p_2 , and internal momenta k and $p_{12} - k$. The diagrams are organized as follows:

- First Diagram:** A sum of two terms. The top term has external momenta p_1 (incoming) and p_1 (outgoing), with internal momenta k and $k - p_1$. The bottom term has external momenta p_2 (incoming) and p_2 (outgoing), with internal momenta k and $k - p_1$. A vertical dashed red line is positioned between the two terms.
- Second Diagram:** A sum of three terms. The top term has external momenta k (incoming) and $k - p_1$ (outgoing), with internal momenta p_1 and $p_{12} - k$. The middle term has external momenta p_1 (incoming) and k (outgoing), with internal momenta k and $k - p_1$. The bottom term has external momenta $k - p_2$ (incoming) and $k - p_2$ (outgoing), with internal momenta $p_{12} - k$ and p_2 . A vertical dashed red line is positioned between the top and middle terms.
- Third Diagram:** A sum of two terms. The top term has external momenta k (incoming) and $k - p_1$ (outgoing), with internal momenta p_1 and $k - p_1$. The bottom term has external momenta $k - p_2$ (incoming) and $k - p_2$ (outgoing), with internal momenta $p_{12} - k$ and p_2 . A vertical dashed red line is positioned between the two terms.

The final result is given by the following integral:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{q\bar{q}} \delta_{q^2}^+(p_{12} - k) \delta_0^+(k)}{p_1^2 p_2^2 (k - p_2)^2 (k - p_1)^2}$$